Computer Practical
Clouds and Orographic Precipitation

Cloud microphysics in a single-column model

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1 Overview

In this practical you will use a single-column model (SCM) of the atmosphere to investigate how cloud microphysics affects precipitation as air is lifted over a hill. The model represents aerosol particles, cloud droplets, rain and ice, and includes processes such as condensation, collision–coalescence, ice nucleation, riming and sedimentation.

The aim is not to reproduce every detail of a real orographic cloud. Instead, you will change one process or environmental condition at a time and use the model output to build a physical explanation for what happens.

Keep a practical record: save each figure using a meaningful filename and write one or two sentences describing the settings and the main result.

By the end of the practical you should be able to:

- explain why cloud-droplet number concentration affects the warm-rain process;
- distinguish warm-rain precipitation from precipitation involving the ice phase;
- explain qualitatively how aerosol particles larger than 0.5 μm influence primary ice in the model;
- describe why a deeper cloud can produce more precipitation;
- recognise that parameterised microphysical processes depend on assumptions as well as on the resolved cloud state.

2 What the model represents

The SCM follows a vertical column of air as it is forced upwards and then downwards over idealised orography. Time in the model can therefore be interpreted as distance travelled across the hill. As the air rises it cools, cloud water forms, and cloud particles can grow into precipitation particles.

Potential temperature is related to temperature and pressure by

$$\theta = T\left(\frac{10^5}{P}\right)^{R_d/c_p}. \tag{1}$$

A generic model variable ϕ is transported vertically according to an equation of the form

$$\frac{\partial \phi}{\partial t} + w\frac{\partial \phi}{\partial z} = S_\phi, \tag{2}$$

where w is vertical air velocity and S_ϕ represents microphysical sources and sinks. For falling hydrometeors the transport speed also includes their sedimentation velocity.

One important growth mechanism is collection. Schematically, the mass growth of a large drop can be written as

$$\frac{dm_i}{dt} \propto D_i^{2+b}\sum_j d_j^3 n_j, \tag{3}$$

so the rate depends on the sizes and concentrations of the particles available for collision. This is why changing droplet number can alter how readily cloud water is converted into rain.

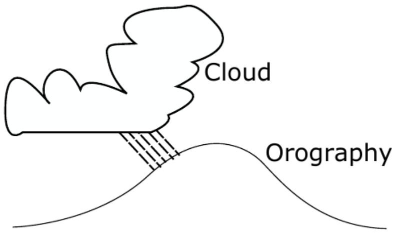

Figure 1: Schematic of the scenario being modelled. Cloud deepens as the air is lifted over the orography and may produce precipitation.

3 Log in, download and compile the code

Use the same SSH/SFTP workflow as in *Tools of the Trade*. On campus, the teaching-server address is `130.88.66.57`; otherwise use the server details supplied for the course. Replace `YOUR_USERNAME` with your own username.

In your SSH window:

```
ssh -Y YOUR_USERNAME@130.88.66.57
```

Open a second terminal window for file transfer:

```
sftp YOUR_USERNAME@130.88.66.57
```

Your password will not be displayed while you type it.

In the SSH window, obtain the model and compile it:

```
git clone https://github.com/EnvModelling/simple-cloud-model
cd simple-cloud-model
make
```

Compilation may take a little while. You normally need to do this only once after downloading the code, unless the source code itself is changed.

File-transfer reminder. Use the SSH window for `nano`, `git`, `make`, `./run.sh` and `python3`. Use the SFTP window for `get`. In SFTP you can rename a downloaded result at the same time:

```
get REMOTE_FILE LOCAL_FILENAME
```

This is useful because every model run otherwise creates a figure with the same name.

4 Inspecting and running the model

Two input files are used in this practical. Open the main model namelist with line numbers displayed:

```
nano -l prac/namelist_prac.in
```

The main parameters used below include `wr_flag` (around line 24), `num_drop` (around line 28), `tsurf` (around line 43), `t_cbase` (around line 45), `t_ctop` (around line 46), and the additional ice-process flags (around lines 20–22). These approximate line numbers are useful with `nano -l`, but they can change when the repository is updated. If a parameter is not where expected, use `Ctrl-W` and search for its name.

Aerosol properties are set in a second file:

```
nano -l prac/namelist.pamm.bam.in
```

The aerosol mode number concentrations are in `n_aer1` (around line 13); the corresponding dry median diameters, widths, molar masses, densities and effective solute factors are grouped immediately below it, around lines 15–19, in `d_aer1`, `sig_aer1`, `molw_core1`, `density_core1` and `nu_core1`. The supplied file contains a pre-configured coarse, dust-like second mode, but its number concentration is set to zero. This lets you switch that mode on later by changing only one number.

Run the SCM from the repository root:

```
./run.sh
```

The output NetCDF file is written to

```
/tmp/YOUR_USERNAME/output.nc
```

Plot it using the supplied Python script:

```
python3 python/scm_plot.py
```

The figure is written to

```
/tmp/YOUR_USERNAME/scm_plot.png
```

Download it in SFTP and give the local copy a useful name, for example:

```
get /tmp/YOUR_USERNAME/scm_plot.png scm_default.png
```

The four panels show cloud-droplet number concentration, cloud-water mass, rain-water mass and ice-crystal number concentration.

To restore just the supplied practical input files after an experiment, use:

```
git restore prac/namelist_prac.in prac/namelist.pamm.bam.in
```

This discards edits to those two files, so save any settings you want to keep in your notes first.

5 Parameters used in the practical

Table 1: Main parameters changed in the experiments. Number concentrations in the namelists are expressed per kg of dry air. Near the surface, values such as `100e6` $\mathrm{kg^{-1}}$ are of order 100 $\mathrm{cm^{-3}}$.

`Parameter`	Supplied value	Meaning
`num_drop`	100e6 $\mathrm{kg^{-1}}$	Initial cloud-droplet number concentration
`wr_flag`	.true.	Switch for collision–coalescence / warm-rain formation
`tsurf`	303.15 K	Surface temperature
`t_cbase`	293.15 K	Cloud-base temperature
`t_ctop`	291.15 K	Cloud-top temperature; together with `t_cbase` controls cloud depth
`n_aer1(2)`	0 $\mathrm{kg^{-1}}$	Number concentration of the pre-configured coarse dust-like mode; zero means that the mode is initially switched off
`d_aer1(2)`	2×10^{-6} m	Dry median diameter of the coarse aerosol mode
`mode1_ice_flag`	0	Experimental secondary-ice mode 1 off
`mode2_ice_flag`	0	Experimental secondary-ice mode 2 off
`coll_breakup_flag1`	0	Ice–ice collisional breakup off

Note. The second aerosol mode is pre-configured as the dust-like mode used in Section 6.3, but its number is zero by default. The default `hm_flag = .true.` enables Hallett–Mossop; the three additional secondary-ice options listed above are initially off.

6 Experiments

For every experiment, save the figure before starting the next run. Unless instructed otherwise, restore the supplied defaults before changing the next parameter.

6.1 Changing cloud-droplet number

Run the supplied default case, plot it and save the figure. Look first at panel (a). The cloud-droplet field decreases as the air passes over the mountain. Compare this with panel (c): as cloud water is converted into sufficiently large drops, rain develops and falls towards the surface. This collision–coalescence pathway is the warm-rain process.

The `num_drop` setting is around line 28 of `prac/namelist_prac.in`. Repeat the experiment with:

Run	Set `num_drop` to	Suggested local filename
A	`10.e6`	`drop_10.png`
B	`100.e6`	`drop_100.png`
C	`1000.e6`	`drop_1000.png`
D	`2000.e6`	`drop_2000.png`

Prediction. Before running all four cases, predict whether increasing droplet number will make collision–coalescence easier or harder when the amount of cloud water is similar.

Exercise. For the default case, where relative to the mountain does rain develop and reach the surface? As cloud-droplet number is reduced or increased, how does the position and amount of rain change? Explain the result in terms of typical droplet size and collision–coalescence.

6.2 Switching off the warm-rain process

Restore the supplied defaults and turn off warm rain (`wr_flag`, around line 24):

```
wr_flag=.false.
```

Run and save the result. Then repeat with

```
num_drop=2000.e6
```

while keeping warm rain switched off.

Exercise. Compare these runs with the corresponding warm-rain cases from Section 6.1. What happens to the persistence of cloud water and to the rain field when collision–coalescence is disabled? Why is the effect especially clear at high droplet number?

6.3 Cooler conditions: ice, secondary ice and INP

Restore the supplied defaults. The temperature settings are around lines 43–48 of `prac/namelist_prac.in`. Now create a much colder cloud by setting

```
tsurf=290.
t_cbase=262.
t_ctop=260.
```

Run the model with `wr_flag=.true.`, plot it and save it as, for example, `cold_warmrain.png`. Then repeat with

```
wr_flag=.false.
```

and save the result as `cold_no_warmrain.png`.

In these runs, examine panel (d) as well as the cloud- and rain-water panels. At these temperatures the cloud is mixed-phase or ice-containing, so precipitation can form through pathways that do not require warm-rain collision–coalescence.

Exercise. What happens to precipitation when warm rain is switched off in the cold case? Is the result the same as in Section 6.2? Use the appearance of ice in panel (d) to explain the difference.

Secondary ice production without the dust perturbation

Keep the cold temperatures, `wr_flag=.false.`, and leave the pre-configured dust-like aerosol mode switched off, so that `n_aer1(2)=0`. The supplied namelist already has `hm_flag=.true.`; now turn on the three additional experimental secondary-ice-production (SIP) options around lines 20–22 of `prac/namelist_prac.in`:

```
mode1_ice_flag=1
mode2_ice_flag=1
coll_breakup_flag1=2
```

Run, plot and save this case as `cold_sip.png`. Option 2 for `coll_breakup_flag1` selects the Phillips-type ice–ice collisional-breakup treatment in the current code.

Compare `cold_sip.png` directly with `cold_no_warmrain.png`. The aerosol population is unchanged between the two runs, so any difference isolates the effect of the additional SIP processes rather than an INP perturbation.

Exercise. How much does the ice-crystal number concentration change when the additional SIP mechanisms are enabled? Does the change in ice number alter the rain or cloud-water fields? Why can secondary ice produce many more ice crystals than would be expected from primary ice nucleation alone?

Adding a pre-configured dust-like coarse aerosol mode

Now switch the three additional SIP flags back to zero so that the dust experiment is a separate controlled comparison:

```
mode1_ice_flag=0
mode2_ice_flag=0
coll_breakup_flag1=0
```

Keep the cold temperatures and `wr_flag=.false.`.

The model's primary-ice calculation uses the DeMott parameterisation. In this implementation, the diagnosed INP concentration depends on temperature and on the number of aerosol particles larger than 0.5 μm. It does *not* identify a particle as mineral dust from its chemical composition.

The second aerosol mode in the supplied `prac/namelist.pamm.bam.in` is already configured as a coarse, weakly hygroscopic, mineral-dust-like mode: $d = 2\,\mu$m, density = 2600 kg m^{-3}, `molw_core1(2)=258.16e-3` kg mol^{-1} and `nu_core1(2)=0.01`. Its number concentration is initially zero. To add the coarse mode, change **only the second number** on the `n_aer1` line (around line 13 of `prac/namelist.pamm.bam.in`) from zero to `1000.e6` kg^{-1}:

```
n_aer1(1:3) = 6000.e6, 1000.e6, 0.1e6,
```

Run, plot and save this case as `cold_dust.png`.

Note. This is a deliberately large perturbation chosen to make the model response visible. Mineral dust is a mixture of minerals and does not have one physically unique molecular weight. Most importantly, it is the increase in the number of coarse particles above 0.5 μm—rather than the chosen `molw_core1` or `nu_core1`—that drives the DeMott INP response in this SCM.

Exercise. Compare `cold_dust.png` with `cold_no_warmrain.png`. How does the ice-crystal number concentration change when the coarse aerosol mode is added? Does the precipitation field also change? Contrast this primary-ice/INP sensitivity with the SIP sensitivity in the previous experiment.

Note. If time permits, you can combine the dust perturbation and the additional SIP switches after you have first examined them separately. Interpreting the two controlled comparisons above is more important than the combined run.

6.4 Deeper cloud

Restore the supplied defaults. Set num_drop (around line 28) and the cloud-base/cloud-top temperatures (around lines 45–46) as follows:

```
num_drop=1000.e6
t_cbase=293.15
t_ctop=288.15
```

so that the cloud-base to cloud-top temperature difference is 5 K rather than the supplied 2 K. Run and plot the model. The cloud is now approximately 1 km deep.

Compare this with a run using `num_drop=1000.e6` and the supplied temperatures `t_cbase=293.15`, `t_ctop=291.15`.

Exercise. How does increasing the cloud depth affect the amount and location of precipitation? Why does giving cloud particles more time and vertical distance in cloud favour precipitation formation?

7 What you should take away

Cloud precipitation is not controlled by a single parameter. In the warm cloud, droplet number affects how efficiently collision–coalescence can produce rain. In the colder cloud, ice-phase pathways allow precipitation to develop even when warm rain is disabled. The primary-ice response is parameterised using temperature and coarse-aerosol number. The additional secondary-ice mechanisms can strongly amplify ice number even without the dust perturbation, while the separate coarse-aerosol experiment shows how primary ice responds to the assumed INP population.

When interpreting any microphysics model, ask the same three questions: which process is explicitly represented, which quantity controls its parameterisation, and which aspects of the real atmosphere are not represented by that parameterisation?